

Cheap Attention, Costly Commitment: Cross-Route Inference in a Dynamic Matching Market

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Abstract

Digital matching markets frequently combine low-cost expressions of short-term interest with a more selective market for long-term commitment. This paper asks when attention in the first route changes choices in the second. Agents face a two-period choice between an available long-term match and a short-term match with a higher-ranked partner. Short-term approaches are generated by endogenous offer decisions and form a Poisson signal of a latent state governing long-term attainability. Agents know payoffs but may use an overly diagnostic likelihood, treating short-term attention as if it were more selective than it is. We characterize the posterior wedge, derive the exact signal histories on which misspecification causes inefficient delay, and embed the resulting acceptance rate in a two-sided participation game. An equilibrium always exists; a global slope restriction gives uniqueness, while local comparative statics identify the participation multiplier. When the restriction fails, a fold theorem gives sufficient conditions for the disappearance of a high-liquidity equilibrium. A route-transparency policy can therefore have both a direct decision effect and an equilibrium participation effect. The mechanism is side-neutral: demographic asymmetries enter as estimable primitives, not assumptions.

JEL codes: C78, D83, D91, J12. **Keywords:** matching markets, misspecified beliefs, online dating, Bayesian learning, search, attention.

1 Introduction

Many digital platforms make an expression of interest nearly costless. That design expands the set of feasible contacts, but it also changes what a contact means. A short-term approach can be profitable for a sender across a wide range of recipients even when long-term commitment would be profitable only for a much narrower set. The two actions then have different selectivity. A recipient who treats their informational content as interchangeable may infer too much about the set of long-term partners they can attain.

This paper isolates that cross-route inference problem. A receiver observes a count of short-term approaches from highly ranked partners and chooses between an available long-term match and a short-term match carrying a possibility of later conversion. The count is generated by one of two latent attainability states. A calibrated receiver uses the true Poisson likelihood. A misspecified receiver uses a likelihood under which short-term attention is more diagnostic of long-term attainability. The behavioral assumption is therefore explicit: attention itself does not create a Bayesian bias. Bias appears only because the receiver applies the wrong statistical model to that attention.

The analysis delivers four results. First, the log-odds belief wedge is affine in the observed count. This produces a transparent cutoff above which the misspecified posterior exceeds the calibrated posterior. Second, the dynamic choice has a closed-form threshold in the perceived

probability of conversion. The combination of the two cutoffs identifies the exact offer counts at which a receiver rejects a long-term match that they would prefer under calibrated beliefs. Third, the acceptance rate enters a two-sided participation game. An equilibrium exists, is unique under a global contraction condition, and has strictly increasing participation and match volume along any stable interior branch. Fourth, when strategic complementarity is stronger, a fold condition generates a genuine liquidity threshold: a small reduction in acceptance can eliminate the high-participation equilibrium.

The distinction from the closest theory is substantive. Antler and Bachi (2022) show that coarse strategic reasoning can produce overoptimistic search, perpetual delay, and market failure. This paper does not claim those conclusions as new. Its object is instead *cross-route inference*: the observed signal is an endogenous count of cheap approaches, the misspecification concerns the likelihood mapping that count into long-term attainability, and platform transparency changes both acceptance and two-sided participation. This delivers history-by-history welfare losses and a route-specific policy comparative static that are absent from a model of coarse depth of reasoning.

The paper deliberately does not assume that one demographic group sends and another receives, that attractiveness has a universal scalar ordering, or that short-term interaction has negligible cost. The roles “proposer” and “receiver” are market roles. Sex, orientation, age, and platform-specific differences can shift priors, arrival rates, payoffs, and costs, but the theory does not determine the signs of those differences.

The model also draws a boundary around its welfare claim. It does not treat disappointment as deadweight loss. Individual welfare falls only on histories where the action chosen under the subjective likelihood has lower payoff under the true data-generating process. “Collapse” refers only to the disappearance of a high-liquidity equilibrium under the explicit fold conditions below.

2 Related literature

The paper connects matching theory, statistical signaling, and equilibrium under misspecification. The distinction between observable actions and the private information behind them is central to signaling theory (Spence, 1973). Here, however, a contact is primarily a statistical signal of the receiver’s market state rather than a costly separating signal chosen to reveal the sender’s type. When short-term contact is cheap and broadly issued, it is naturally pooling rather than separating.

The behavioral friction is related to cursed equilibrium, in which agents underappreciate the relationship between others’ information and actions (Eyster and Rabin, 2005), and to Berk–Nash equilibrium, which gives a learning foundation for optimal behavior under a misspecified model (Esponda and Pouzo, 2016). The baseline below is a finite subjective Bayesian model; the repeated-learning interpretation is discussed in Section 8.

Classical matching work establishes stability and sorting under explicit preference and surplus conditions (Gale and Shapley, 1962; Becker, 1973). Search models of marriage show how frictions and reservation strategies alter sorting (Burdett and Coles, 1997); Chade et al. (2017) survey the broader search-and-matching literature. The two papers closest in environment are Antler and Bachi (2022), where coarse reasoning produces overoptimistic search, and Antler et al. (ming), where partners rationally learn compatibility through dating.

Empirical work using online-dating interactions finds structured preferences, sorting, and aspirational pursuit (Hitsch et al., 2010; Bruch and Newman, 2018). Field evidence also shows that information about the number of agents on each side changes participation and selectivity (Fong, 2024), while a job-fair experiment documents distorted beliefs and changed search behavior on both sides of a matching market (Abebe et al., 2025). These findings motivate a model in which

Table 1: Closest literature and contribution boundary

Study	Central object	Distinction of this paper
Antler and Bachi (2022)	Coarse strategic reasoning in two-sided search; excessive search and possible market failure.	Endogenous attention counts, a route-specific likelihood error, exact distorted histories, and transparency.
Antler et al. (ming)	Correct Bayesian learning about match-specific compatibility while dating.	Misspecified inference from one route about attainability in another route.
Lee and Niederle (2015)	Scarce “rose” messages as deliberate preference signals.	Abundant approaches are statistical signals generated by cutoff offer choices, not separating messages.
Fong (2024)	Experimental effects of perceived market size and competition on participation and selectivity.	A mechanism linking attention information to two-sided participation and a liquidity threshold.
Andrew and Adams (2025)	Identification of subjective marriage-market beliefs from dynamic hypothetical choices.	A structural source of belief error and its equilibrium feedback, rather than a belief-elicitation method.

an approach, an inference, participation, and an eventual match are distinct objects.

Reference-dependent preferences could amplify post-match dissatisfaction (Kőszegi and Rabin, 2006), but they are not needed for the baseline mechanism and are kept outside the main model. This separation matters empirically: distorted beliefs change choices before matching, whereas reference dependence changes experienced utility after expectations have formed.

3 Environment

3.1 Agents, dates, and routes

There are two dates, $t \in \{1, 2\}$, and two sides of a matching market. We focus on the decision of a representative receiver. The receiver can form a long-term match with an average-ranked proposer, denoted A , or accept a one-period short-term match with a higher-ranked proposer, denoted H . These labels refer to the receiver’s payoff ranking, not to immutable biological types.

At date 1 the receiver faces a stylized menu. Accepting the A long-term offer creates a match in both periods. Choosing the H short-term route produces a date-1 payoff and may convert to a long-term H match at date 2. If conversion does not occur, an A long-term opportunity is available at date 2 with probability $r \in [0, 1]$. The menu is best interpreted as the decision problem conditional on encountering both routes; Section 7 describes how route availability can be estimated rather than normalized.

Let $L_A > 0$ be the receiver’s net per-period value from the A long-term match, $L_H > L_A$ the date-2 value of a converted H match, and $S_H \in \mathbb{R}$ the net date-1 value of the short-term interaction. These values include time, health, emotional, search, and opportunity costs. The discount factor is $\delta \in (0, 1]$. Thus the value of accepting A at date 1 is

$$V_L = (1 + \delta)L_A. \quad (1)$$

If the probability of conversion is q , the value of the short-term route is

$$V_S(q) = S_H + \delta [qL_H + (1 - q)rL_A]. \quad (2)$$

3.2 Latent attainability and attention

The receiver is uncertain about a route-specific market state $d \in \{0, 1\}$. The state summarizes how likely highly ranked short-term partners are to find a long-term match with this receiver mutually acceptable. The prior is

$$\mathbb{P}(d = 1) = p \in (0, 1).$$

This is uncertainty about equilibrium long-term attainability, not uncertainty about one's directly observed physical attributes or intrinsic worth.

During a fixed exposure window, a Poisson mass $m_H > 0$ of potential H proposers encounters the receiver. Each draws a private cost k from continuous cdf K_S and obtains state-dependent expected gross value v_d^S from making a short-term approach. A proposer sends the approach if and only if $k \leq v_d^S$. Independent thinning of a Poisson process therefore gives

$$N \mid d \sim \text{Poisson}(\lambda_d), \quad \lambda_d = m_H K_S(v_d^S). \quad (3)$$

We assume $0 < K_S(v_0^S) < K_S(v_1^S)$, hence $0 < \lambda_0 < \lambda_1$. Thus the count distribution is induced by optimal sender cutoffs rather than imposed directly. When $v_1^S - v_0^S$ is small, or K_S is locally flat, the true likelihood ratio λ_1/λ_0 is close to one: cheap short-term attention is weakly selective even though offer costs are positive. The calibrated posterior is

$$\pi(n) = \mathbb{P}(d = 1 \mid N = n) = \frac{pe^{-\lambda_1}\lambda_1^n}{pe^{-\lambda_1}\lambda_1^n + (1-p)e^{-\lambda_0}\lambda_0^n}. \quad (4)$$

Its log odds satisfy

$$\text{logit } \pi(n) = \text{logit } p + n \log \frac{\lambda_1}{\lambda_0} - (\lambda_1 - \lambda_0). \quad (5)$$

Conversion occurs with probability γ_d in state d , where

$$0 \leq \gamma_0 < \gamma_1 \leq 1. \quad (6)$$

The objective conversion probability conditional on the observed history is

$$\gamma(n) = \gamma_0 + (\gamma_1 - \gamma_0)\pi(n). \quad (7)$$

Because $\lambda_1 > \lambda_0$, both $\pi(n)$ and $\gamma(n)$ are strictly increasing in n .

3.3 Cross-route misspecification

The receiver knows payoffs and the state-dependent conversion probabilities but may use the wrong likelihood for the offer count. Specifically, the subjective model is

$$N \mid d \overset{\text{subj}}{\sim} \text{Poisson}(\tilde{\lambda}_d), \quad 0 < \tilde{\lambda}_0 < \tilde{\lambda}_1. \quad (8)$$

The intended interpretation is that the receiver treats short-term attention as if it had been generated by a more selective, commitment-oriented process. The subjective posterior and conversion probability are

$$\tilde{\pi}(n) = \frac{pe^{-\tilde{\lambda}_1}\tilde{\lambda}_1^n}{pe^{-\tilde{\lambda}_1}\tilde{\lambda}_1^n + (1-p)e^{-\tilde{\lambda}_0}\tilde{\lambda}_0^n}, \quad (9)$$

$$\tilde{\gamma}(n) = \gamma_0 + (\gamma_1 - \gamma_0)\tilde{\pi}(n). \quad (10)$$

Assumption 1 (Overdiagnostic subjective model). *The subjective likelihood ratio is more responsive to the count than the true likelihood ratio:*

$$\frac{\tilde{\lambda}_1}{\tilde{\lambda}_0} > \frac{\lambda_1}{\lambda_0}.$$

Assumption 1 does not imply that the subjective posterior is above the true posterior at every count. Low counts can be interpreted too pessimistically. The next result identifies the crossing exactly.

Proposition 1 (Posterior wedge). *Under Assumption 1, define*

$$\begin{aligned}\omega &= \log \frac{\tilde{\lambda}_1/\tilde{\lambda}_0}{\lambda_1/\lambda_0} > 0, \\ \eta &= (\tilde{\lambda}_1 - \tilde{\lambda}_0) - (\lambda_1 - \lambda_0).\end{aligned}$$

Then

$$\text{logit } \tilde{\pi}(n) - \text{logit } \pi(n) = n\omega - \eta. \quad (11)$$

Consequently, $\tilde{\pi}(n) > \pi(n)$ and $\tilde{\gamma}(n) > \gamma(n)$ if and only if $n > \eta/\omega$.

The proposition makes the behavioral content falsifiable. High attention causes overoptimism only when the receiver's assumed likelihood is sufficiently more diagnostic than the true one. Under correct Bayesian updating, $\tilde{\lambda}_d = \lambda_d$ and the wedge is identically zero.

4 Dynamic choice and equilibrium

By (1)–(2), the receiver prefers the short-term route when the perceived conversion probability exceeds

$$\gamma^* = \frac{(1 + \delta)L_A - S_H - \delta r L_A}{\delta(L_H - r L_A)}. \quad (12)$$

Assumption 2 (Interior choice). $L_H > r L_A$ and $\gamma^* \in (\gamma_0, \gamma_1)$.

The first inequality makes conversion valuable relative to returning to the average market. The second rules out cases in which one route is optimal for every possible belief. Let

$$\pi^* = \frac{\gamma^* - \gamma_0}{\gamma_1 - \gamma_0} \in (0, 1). \quad (13)$$

We use the following finite subjective-equilibrium concept.

Definition 1 (Subjective Bayesian equilibrium). *A subjective Bayesian equilibrium consists of a receiver strategy $a(n)$, subjective beliefs $\tilde{\pi}(n)$, and an objective outcome distribution such that:*

- (i) $a(n)$ maximizes (1) and (2) under $\tilde{\pi}(n)$;
- (ii) beliefs are generated by the subjective model (8); and
- (iii) objective outcome frequencies are generated by the true model (3) and the strategy $a(n)$.

The calibrated equilibrium is the special case $\tilde{\lambda}_d = \lambda_d$.

The receiver's subjective and calibrated count thresholds are

$$\tilde{n} = \min\{n \in \mathbb{N}_0 : \tilde{\pi}(n) \geq \pi^*\}, \quad (14)$$

$$n^0 = \min\{n \in \mathbb{N}_0 : \pi(n) \geq \pi^*\}. \quad (15)$$

Away from knife-edge equalities, these have closed forms

$$n^0 = \left\lceil \frac{\text{logit } \pi^* - \text{logit } p + (\lambda_1 - \lambda_0)}{\log(\lambda_1/\lambda_0)} \right\rceil, \quad (16)$$

$$\tilde{n} = \left\lceil \frac{\text{logit } \pi^* - \text{logit } p + (\tilde{\lambda}_1 - \tilde{\lambda}_0)}{\log(\tilde{\lambda}_1/\tilde{\lambda}_0)} \right\rceil. \quad (17)$$

Proposition 2 (Cross-route distortion). *Under Assumption 2, the calibrated receiver chooses S if and only if $n \geq n^0$, while the misspecified receiver chooses S if and only if $n \geq \tilde{n}$. If $\tilde{n} < n^0$, then exactly the histories*

$$\mathcal{D} = \{\tilde{n}, \tilde{n} + 1, \dots, n^0 - 1\} \quad (18)$$

produce inefficient delay. For each $n \in \mathcal{D}$, the receiver's material welfare loss is

$$\ell(n) = V_L - V_S(\gamma(n)) = \delta(L_H - rL_A) [\gamma^* - \gamma(n)] > 0. \quad (19)$$

No distortion occurs if $\tilde{n} = n^0$.

The result distinguishes preference from error. A receiver may rationally value a short-term match, and a high count may rationally predict conversion. The model calls the choice inefficient only on histories where the receiver would choose the long-term route under the true likelihood but chooses the short-term route under the subjective likelihood.

Let

$$f(n) = (1 - p)e^{-\lambda_0} \frac{\lambda_0^n}{n!} + pe^{-\lambda_1} \frac{\lambda_1^n}{n!} \quad (20)$$

be the unconditional count distribution. Expected receiver welfare loss is

$$\mathcal{L} = \sum_{n \in \mathcal{D}} f(n) \ell(n). \quad (21)$$

It is strictly positive whenever $\mathcal{D}$ is nonempty, because every finite Poisson count has positive probability.

5 Two-sided participation and liquidity

We now embed the receiver's decision in a two-sided participation game. Let n_c denote the count at which a receiver switches away from the long-term route. The probability that an encountered long-term proposal is accepted is

$$\alpha(n_c) = \mathbb{P}(N < n_c) = \sum_{n=0}^{n_c-1} f(n). \quad (22)$$

Hence $\alpha^0 = \alpha(n^0)$ under calibrated inference and $\tilde{\alpha} = \alpha(\tilde{n})$ under the subjective model. If $\tilde{n} < n^0$, every count has positive probability and therefore $\tilde{\alpha} < \alpha^0$.

There is a unit mass on each long-term side, proposer side P and receiver side R . An agent on side $g \in \{P, R\}$ draws an idiosyncratic net participation cost c_g from a continuously differentiable cdf F_g on $\mathbb{R}$, with density f_g . Allowing $F_g(0) > 0$ accommodates agents with a positive baseline value of participation and prevents zero activity from being imposed by construction. If masses x_P and x_R participate, potential meetings are

$$M(x_P, x_R) = \mu x_P x_R, \quad \mu \in (0, 1]. \quad (23)$$

A potential meeting succeeds with probability α . A successful match gives side g benefit $b_g > 0$; write $B_g = \mu b_g$. Thus the participation cutoffs are $B_P \alpha x_R$ and $B_R \alpha x_P$.

Definition 2 (Participation equilibrium). *Given $\alpha \in (0, 1]$, a participation equilibrium is a pair of participation masses in $[0, 1]^2$ satisfying*

$$x_P = F_P(B_P \alpha x_R), \quad (24)$$

$$x_R = F_R(B_R \alpha x_P). \quad (25)$$

Successful long-term match volume is

$$Q(\alpha) = \mu \alpha x_P x_R. \quad (26)$$

Proposition 3 (Existence and global uniqueness). *For every $\alpha \in (0, 1]$, a participation equilibrium exists. If the densities are bounded by $\bar{f}_P$ and $\bar{f}_R$ and*

$$\alpha^2 B_P B_R \bar{f}_P \bar{f}_R < 1, \quad (27)$$

the equilibrium is unique.

Existence does not depend on a contraction or on symmetry. The uniqueness condition has a direct interpretation: the product of the two maximal participation responses must be smaller than one. It separates ordinary market thinning from the strong complementarity required for multiple liquidity equilibria.

Proposition 4 (Participation multiplier). *Consider an interior equilibrium at which both densities are positive. Define*

$$\begin{aligned} z_P &= B_P \alpha x_R, & z_R &= B_R \alpha x_P, \\ \kappa_P &= \alpha B_P f_P(z_P), & \kappa_R &= \alpha B_R f_R(z_R), \\ d_P &= B_P x_R f_P(z_P), & d_R &= B_R x_P f_R(z_R). \end{aligned}$$

If $\kappa_P \kappa_R < 1$, the equilibrium is locally unique and

$$\frac{dx_P}{d\alpha} = \frac{d_P + \kappa_P d_R}{1 - \kappa_P \kappa_R} > 0, \quad (28)$$

$$\frac{dx_R}{d\alpha} = \frac{d_R + \kappa_R d_P}{1 - \kappa_P \kappa_R} > 0. \quad (29)$$

Consequently $dQ/d\alpha > 0$ along the stable branch.

The direct effect of a lower acceptance rate is amplified because exit on either side lowers the return to participation on the other. Equations (28)–(29) quantify that multiplier instead of assigning it an arbitrary reduced-form magnitude.

For a genuine tipping result, substitute (25) into (24) and define

$$H(x; \alpha) = F_P(B_P \alpha F_R(B_R \alpha x)), \quad G(x; \alpha) = H(x; \alpha) - x. \quad (30)$$

Proposition 5 (Liquidity fold). *Suppose F_P and F_R are twice continuously differentiable near $(x^*, \alpha^*) \in (0, 1)^2$ and*

$$G(x^*; \alpha^*) = 0, \quad H_x(x^*; \alpha^*) = 1, \quad H_{xx}(x^*; \alpha^*) < 0, \quad H_\alpha(x^*; \alpha^*) > 0. \quad (31)$$

Then there are neighborhoods of x^ and α^* such that: for $\alpha > \alpha^*$ there are exactly two nearby equilibria; for $\alpha < \alpha^*$ there is no nearby equilibrium. The lower nearby fixed point has $H_x > 1$ and is unstable under best-response iteration; the upper fixed point has $0 \leq H_x < 1$ and is locally stable.*

Proposition 5 is a local saddle-node theorem, not a claim that every market has multiple equilibria. If a separate low stable equilibrium persists, reducing α through α^* removes the high-liquidity branch and best-response dynamics select the remaining low-liquidity state. This is the precise sense in which cross-route inference can tip a market.

An information policy that reveals the true route-specific arrival rates sets $\tilde{\lambda}_d = \lambda_d$. In the decision problem it eliminates $\mathcal{D}$, raises acceptance from $\tilde{\alpha}$ to α^0 , and removes $\mathcal{L}$. In the participation game it raises activity along a stable branch and can restore a high-liquidity equilibrium across a fold. Whether a real label changes sender behavior or count composition is an empirical question, so these policy claims are conditional on holding the true offer technology fixed.

6 Numerical illustration

This section illustrates magnitudes; it is not a calibration. Consider the parameters in Table 2. Short-term attention is only weakly diagnostic under the true model, with arrival rates 2 and 3, but the receiver behaves as if the rates were 1 and 4.

Table 2: Illustrative parameter values

p	λ_0	λ_1	$\tilde{\lambda}_0$	$\tilde{\lambda}_1$	γ_0	γ_1	δ	L_A	L_H	S_H
0.30	2	3	1	4	0.05	0.45	0.90	1.00	1.60	1.10

Note: $r = 0.50$. All payoff values are net and normalized.

Equation (12) gives $\gamma^* = 0.3535$ and $\pi^* = 0.7588$. The calibrated receiver chooses the short-term route beginning at $n^0 = 8$, while the misspecified receiver does so beginning at $\tilde{n} = 4$. Hence $\mathcal{D} = \{4, 5, 6, 7\}$. Table 3 shows the relevant conversion beliefs.

Table 3: Conversion beliefs around the decision threshold

n	$\pi(n)$	$\tilde{\pi}(n)$	$\gamma(n)$	$\tilde{\gamma}(n)$
3	0.347	0.577	0.189	0.281
4	0.444	0.845	0.228	0.388
5	0.545	0.956	0.268	0.432
6	0.642	0.989	0.307	0.445
7	0.729	0.997	0.342	0.449
8	0.802	0.999	0.371	0.450

The distortion set has unconditional probability 0.2015, and expected receiver loss is 0.0201 in normalized payoff units. The calibrated and subjective long-term acceptance probabilities are 0.9957 and 0.7942.

To illustrate the two-sided feedback, set $B_P = B_R = \mu = 1$ and use the symmetric net-cost cdf

$$F(z) = \{1 + \exp[-10(z - 0.55)]\}^{-1}.$$

Symmetric equilibria solve $x = F(\alpha x)$. The high-liquidity fold occurs at $(x^*, \alpha^*) = (0.8639, 0.8506)$. At calibrated acceptance the fixed points are 0.0042, 0.5882, and 0.9870; the first and last are stable. Starting on the high branch, the fall from 0.9957 to 0.7942 crosses the fold, leaving only the low fixed point 0.0042. Long-term match volume falls from 0.9699 on the calibrated high branch to 0.000014 in the remaining subjective equilibrium. These values demonstrate the theorem’s mechanism, not an empirical calibration or a generic prediction of collapse. The replication code computes the fixed points and checks their residuals.

7 Empirical content

The mechanism is testable only with data that distinguish attention from commitment. A useful dataset would record profile exposure, route or stated intent, approaches, replies, meetings, long-term proposals, and realized conversions. Measuring exposure is essential: raw counts confound recipient attributes with the platform’s ranking and recommendation algorithm.

The model generates five predictions.

1. Holding estimated long-term type fixed, an exogenous increase in short-term attention raises perceived long-term attainability when users are not shown route-specific base rates.
2. The effect is concentrated around the choice cutoff: receivers far below or above γ^* do not change actions.
3. Revealing route labels and empirical conversion rates reduces the effect of short-term offer counts on long-term reservation behavior.
4. A belief-induced decline in long-term acceptance reduces participation on both long-term sides; the response becomes large where the estimated product of best-response slopes approaches one.
5. The relevant sufficient statistic is the likelihood ratio of attention across latent attainability states, not the raw number of approaches.

A clean design would randomize additional profile exposure or a batch of clearly labeled, low-commitment signals and elicit beliefs before and after the shock. The outcome is subsequent acceptance of independently generated long-term opportunities. A second treatment would reveal the platform’s route-specific base rates. The key estimand is the interaction between the attention shock and transparency treatment.

Observational estimates face three major threats. First, recipients who receive more attention may truly have higher long-term attainability. Second, platform algorithms jointly determine exposure and partner composition. Third, stated intent may not equal actual commitment. Randomized exposure, sender fixed effects, experimentally varied labels, and observed conversions are therefore more informative than cross-sectional correlations between message counts and later outcomes.

The model is intentionally side-neutral. Empirical work may estimate λ_d , $\hat{\lambda}_d$, γ_d , payoffs, and entry costs separately by sex, age, orientation, or platform. A claim that the mechanism is stronger for one group requires those estimates; it is not a theorem of the model.

8 Extensions and scope

Strategic signaling. The baseline treats an offer as a statistical signal. A genuine signaling extension would give proposers a private commitment cost and let them choose a costly message or bond. A separating equilibrium would require single crossing. If both commitment types can send the same cheap short-term offer, the offer is pooling and the posterior over commitment remains the prior.

Repeated learning. With many episodes, agents may estimate the wrong model that best approximates observed outcomes within a restricted subjective family. A Berk–Nash formulation would replace the fixed pair $(\tilde{\lambda}_0, \tilde{\lambda}_1)$ with the Kullback–Leibler minimizer inside that family. Selective feedback is likely important: agents observe conversion only for short-term offers they accept.

Reference dependence. Past attention may enter a reference point even when beliefs are calibrated. That mechanism should be modeled as a separate state variable, following the logic of expectations-based reference dependence, rather than inserted into the Bayesian posterior. It can affect experienced match utility without implying that accepting a positive-surplus match is irrational.

Investment. If agents choose costly appearance or relationship investment x , effective quality can be written $q = b + g(x)$ with fixed endowment b , concave technology g , and convex cost K . The private first-order condition is $K'(x) = g'(x)\mathbb{E}[V_q]$. When returns are mainly ordinal, simultaneous investment can become a positional arms race and need not change relative tiers. No universal physical target follows from the theory.

9 Conclusion

Cheap short-term attention and costly long-term commitment need not carry the same information. A calibrated Bayesian accounts for that difference, so a large volume of attention alone cannot generate systematic overconfidence. The mechanism in this paper requires a precise error: the receiver uses a likelihood under which short-term approaches are more selective about long-term attainability than they truly are.

That error has sharp consequences in the two-period model. It moves the count threshold at which a receiver chooses the short-term route, creates an exactly characterized set of inefficient-delay histories, and lowers long-term acceptance. In a two-sided participation game, an equilibrium always exists, the stable comparative statics amplify the direct acceptance effect, and an explicit fold condition identifies when a high-liquidity equilibrium disappears. These are conditional, falsifiable claims. Stronger statements about demographic asymmetry, dissatisfaction, or household stability require additional primitives and separate evidence.

A Proofs

Proof of Proposition 1. Bayes' rule gives

$$\text{logit } \pi(n) = \text{logit } p + n \log(\lambda_1/\lambda_0) - (\lambda_1 - \lambda_0)$$

and

$$\text{logit } \tilde{\pi}(n) = \text{logit } p + n \log(\tilde{\lambda}_1/\tilde{\lambda}_0) - (\tilde{\lambda}_1 - \tilde{\lambda}_0).$$

Subtracting produces (11). Assumption 1 implies $\omega > 0$, so the difference is positive exactly when $n > \eta/\omega$. The logit function is strictly increasing. Hence the same inequality orders $\tilde{\pi}(n)$ and $\pi(n)$. Finally, $\tilde{\gamma}(n) - \gamma(n) = (\gamma_1 - \gamma_0)[\tilde{\pi}(n) - \pi(n)]$, and $\gamma_1 > \gamma_0$ completes the proof. $\square$

Proof of Proposition 2. Subtracting (1) from (2) yields

$$V_S(q) - V_L = \delta(L_H - rL_A)(q - \gamma^*).$$

By Assumption 2, the coefficient on $q - \gamma^*$ is positive. Thus a receiver chooses S exactly when the relevant conversion belief is at least γ^* . Equations (7), (10), and (13) imply that this is equivalent to the relevant posterior being at least π^* . Both posteriors are strictly increasing in n , which establishes the threshold strategies in (14)–(15).

If $\tilde{n} < n^0$, a subjective receiver chooses S and a calibrated receiver chooses L exactly when $\tilde{n} \leq n < n^0$. On those histories, $\gamma(n) < \gamma^*$, so evaluating the subjective action under the true model gives

$$V_L - V_S(\gamma(n)) = \delta(L_H - rL_A)[\gamma^* - \gamma(n)] > 0.$$

If the thresholds coincide, the two strategies coincide for every count. $\square$

Proof of Proposition 3. For fixed α , define

$$T_\alpha(x_P, x_R) = (F_P(B_P\alpha x_R), F_R(B_R\alpha x_P)).$$

The map is continuous and maps the nonempty compact convex set $[0, 1]^2$ into itself. Brouwer's fixed-point theorem gives existence.

For uniqueness, let $x = (x_P, x_R)$ and $y = (y_P, y_R)$ be equilibria. The bounded density assumption and the mean-value theorem imply

$$\begin{aligned} |x_P - y_P| &\leq B_P\alpha\bar{f}_P|x_R - y_R|, \\ |x_R - y_R| &\leq B_R\alpha\bar{f}_R|x_P - y_P|. \end{aligned}$$

Combining the inequalities yields

$$|x_P - y_P| \leq \alpha^2 B_P B_R \bar{f}_P \bar{f}_R |x_P - y_P|.$$

Under (27), this is possible only if $x_P = y_P$; the second inequality then implies $x_R = y_R$. $\square$

Proof of Proposition 4. Differentiating (24)–(25) at an interior equilibrium gives

$$\begin{aligned} dx_P &= d_P d\alpha + \kappa_P dx_R, \\ dx_R &= d_R d\alpha + \kappa_R dx_P. \end{aligned}$$

The Jacobian determinant is $1 - \kappa_P \kappa_R > 0$, so the implicit-function theorem gives local uniqueness and solving the system gives (28)–(29). All terms in their numerators are positive. Finally,

$$\frac{dQ}{d\alpha} = \mu \left[x_P x_R + \alpha x_R \frac{dx_P}{d\alpha} + \alpha x_P \frac{dx_R}{d\alpha} \right] > 0.$$

$\square$

Proof of Proposition 5. At the candidate point, $G = G_x = 0$, $G_{xx} < 0$, and $G_\alpha > 0$. Because $G_{xx} \neq 0$, the implicit-function theorem applied to $G_x = 0$ gives a unique continuously differentiable critical point $x_c(\alpha)$ near x^* . Shrinking the neighborhood if necessary, $G_{xx} < 0$ throughout, so this point is a strict local maximum. Define $m(\alpha) = G(x_c(\alpha); \alpha)$. The envelope identity gives

$$m'(\alpha^*) = G_\alpha(x^*; \alpha^*) > 0.$$

Thus $m(\alpha)$ is negative immediately below α^* and positive immediately above it. At α^* , strict concavity makes G negative at two sufficiently close boundary points on either side of x^* . By continuity the same is true for nearby α . Hence, below α^* the local maximum is negative and there is no nearby zero; above it, the maximum is positive and the intermediate-value theorem gives one zero on each side. Strict concavity makes these zeros unique in the neighborhood.

At the lower zero $G_x > 0$, so $H_x > 1$; at the upper zero $G_x < 0$, so $H_x < 1$. Because cdf derivatives and all scale parameters are nonnegative, $H_x \geq 0$. The usual one-dimensional fixed-point criterion therefore gives instability of the lower zero and local stability of the upper zero. $\square$

B Verification and reproducibility

The replication archive contains two reproducible checks on the derivations. First, `replicate.py` recomputes every reported number, locates all fixed points by bracketed bisection, and asserts posterior, cutoff, fixed-point, and stability residuals. Second, `formal/ProofAudit.lean` is checked by the Lean kernel against Mathlib. It formalizes the dynamic choice-gap identity, the sign of the individual welfare loss, the contraction inequality used for uniqueness, and the product monotonicity behind match-volume comparisons. The local fold proof remains an analytic argument in the text; the code verifies its numerical instance but is not a formalization of the implicit-function theorem.

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